Technology Stack & Workflow

- **Python:** For Data cleaning, Preprocessing, and Feature Engineering.

Part 1: Python Data cleaning, Preprocessing, and Feature Engineering.

Phase 1: Data Loading & Cleaning

- **Data Acquisition:** Successfully loaded the HR dataset using the Pandas, Numpy, Seaborn, Matplotlib library.
- **Handling Redundant Data:** Identified and dropped columns like EmployeeCount, Over18, and StandardHours that do not add value to the analysis.
- **Integrity Check:** Performed a missing value analysis to ensure the dataset is clean and Complete.

> Users > DELL > Desktop > End-to-End HR Data Pipeline > *?* hr_data_preprocessing.py > ...

```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as pit
import warnings

# Suppress warnings for clean output
warnings.filterwarnings('ignore')

#-----------------------------------------------------------
# Phase 1: Data Loading
#-----------------------------------------------------------
try:
df = pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv¹)
print("Data successfully loaded.")
except FileMotFoundError:
print("Error: The source file was not found.")

#-----------------------------------------------------------
# Phase 2: Data Cleaning & Preprocessing
#-----------------------------------------------------------
# Dropping redundant columns that do not add value to the analysis
cols_to_drop = [¹ EmployeeCount', 'OverlS¹., ' StandardHours ', ¹ EmployeeNumber¹ ]
df.drop(columns=cols_to_drop, inplace=True)

# Checking for missing values
missing_data = df.isnull().sum().sum()
print(f"Total Missing Values: {missing_data}")
```

Phase 2: Feature Engineer & Logic

Implementation

Income Categorization: Segmented raw MonthlyIncome into 'Low',

'Medium', and 'High' groups for better analysis.

Predictive Risk Scoring: Developed a custom Risk_Score attribute.

Logic Implementation:

High Risk: Low satisfaction (< 2) + Overtime.

Medium Risk: Low satisfaction but no overtime.

Value Addition: Transformed raw numbers into actionable categorical

insights.# ; : :

```
# Phase 3: Feature Engineering

# Categorizing Monthly Income into Low, Medium, and High
groups
df[1Income_Group#'] = pd.cut(df['MonthlyIncome'],

           bins=[0, 5000,    10000, 25000],
           labels=[* Low", 'Medium', 'High'])

#    Creating a Risk Factor column based on lob Satisfaction and Overtime

#    High Risk = Low Satisfaction AND Working Overtime

df[1Risk_Score'] = 'Low'

df.loc[(df['lobSatisfaction'] <= 2) & (df['OverTime'] == 'Yes'), 'Risk Score'] = 'High'

df.loc[(df['lobSatisfaction'] <= 2) & (df['OverTime'] == 'No'), 'Risk_Score’] = 'Medium
```

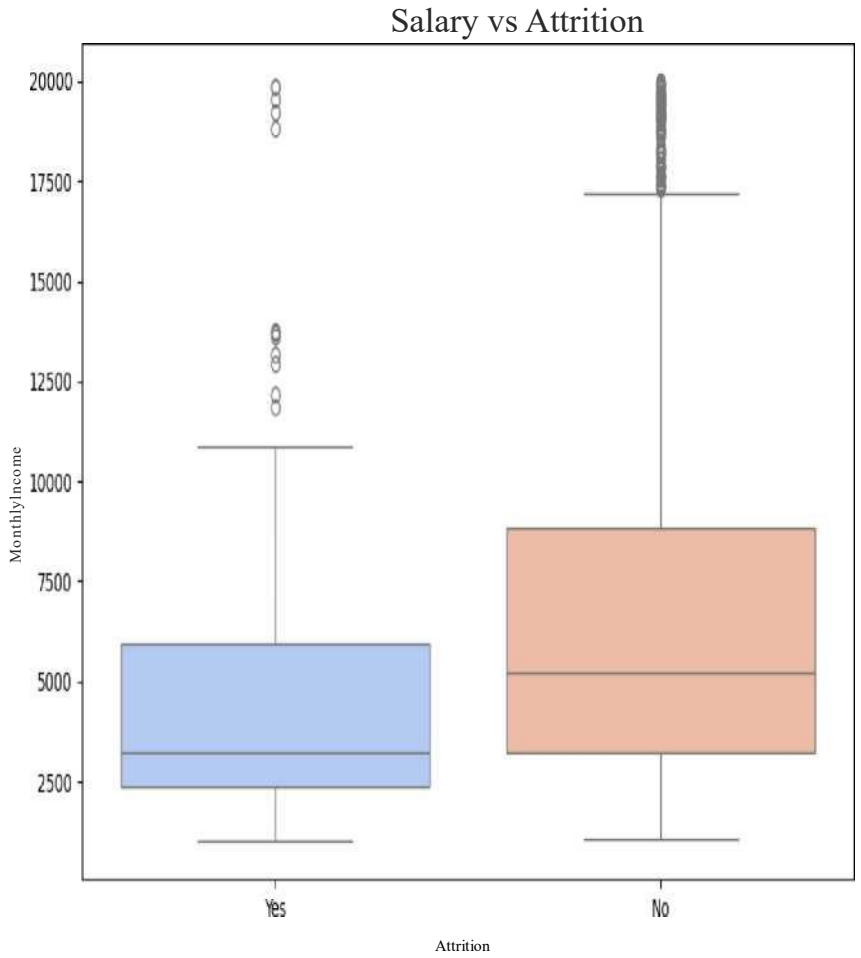
Phase 3: Exploratory Data Analysis (EDA) - Statistical Visualizations

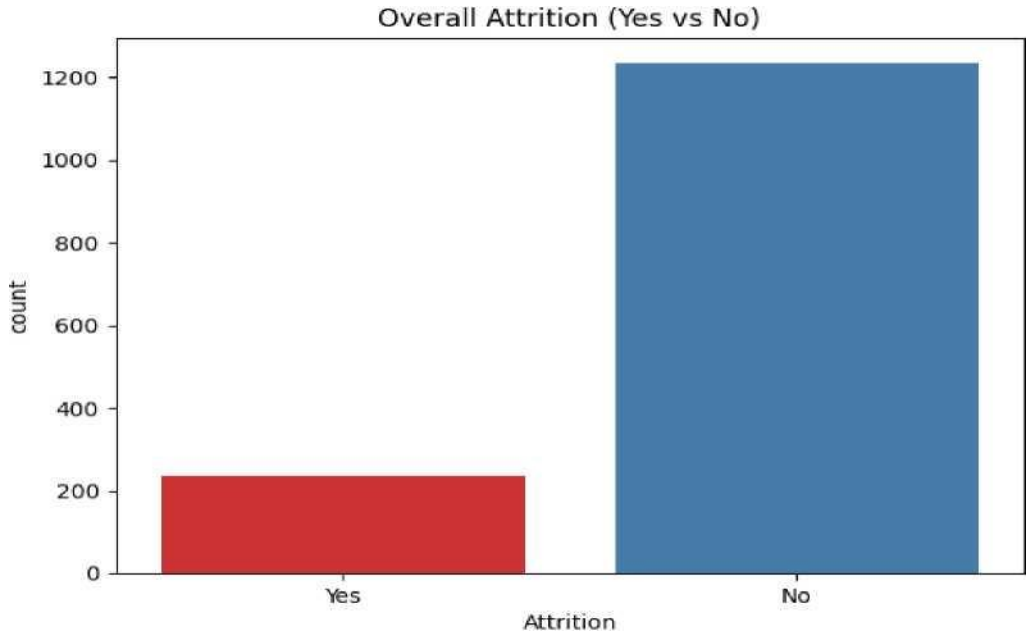
- **Automated Visualization:** Leveraged Matplotlib and Seaborn libraries to generate insightful statistical plots.
- **Attrition Distribution:** Analyzed the overall turnover ratio to understand the balance between current and former employees.
- **Departmental Analysis:** Visualized attrition across different departments to pinpoint high-turnover areas (e.g., Sales vs. R&D).
- **Income Impact:** Utilized Box Plots to study the correlation between Monthly Income and employee attrition.
- **Exporting Insights:** Automated the process of saving visualizations as high-quality PNG files for reporting and documentation.

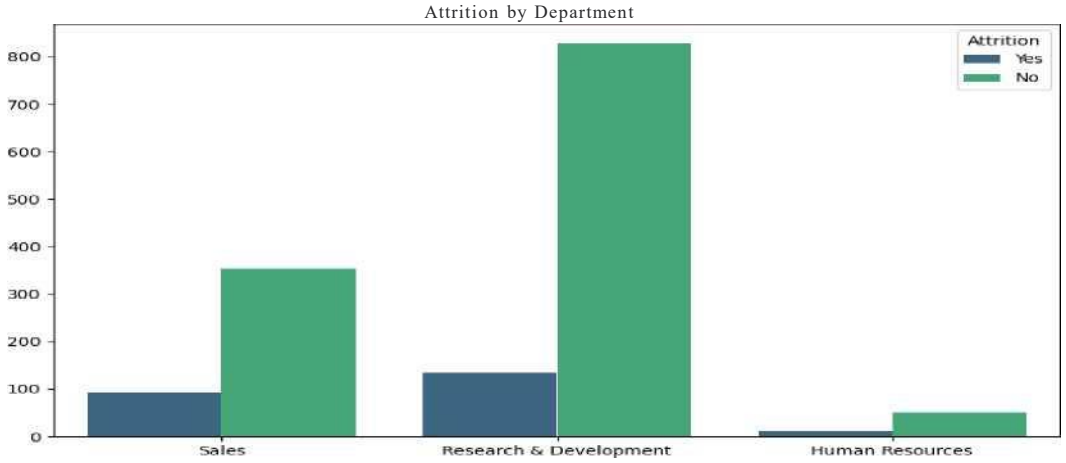
```
i #-----------------------------------------------
>   # Phase 4: Exploratory Data Analysis (Visualizations)
i #-----------------------------------------------
     print("Generating visualizations...")
i
>   # Set aesthetic style of the plots
     sns. set_theme(style= "whitegrid")

/   #1. Distribution of Attrition
     pit.figure(figsize=(Sj 5))
l   sns.countplot(data=dfj x=1 Attrition', hue= Attrition'3 palette—1 Set11, legend=False)
i pit.title('Distribution of Employee Attrition')
>   pit.savefig(Tattrition_distribution.png')
     pit.close()
i
>   #2. Attrition by Department
     pit.figure(figsize=(10, 6))
. sns.countplot(data=dfj x=1 Department', hue='Attrition1 palette=1viridis1)
/ pit.title('Employee Attrition Analysis by Department')
! pit.savefig(T attritionbydepartment.png')
l plt.close()
i
>   #3. Income Levels vs Attrition
     pit.figure(figsize=(10, 6))
     sns. boxplot(data=df,  x='AttritionTj  y='MonthlyIncome's hue='Attrition1, palette='coolwarm', legend=False)
i pit.title('Impact of Monthly Income on Attrition1)
) pit.savefig(1incomevsattrition.png')
. plt.close()
```

Graph:


Salary vs Attrition
20000
17500
15000
12500
10000
7500
5000
2500
Monthlyincome
Yes
No
Attrition


Overall Attrition (Yes vs No)
1200
1000
800
600
400
200
0
count
Yes
No
Attrition


Attrition by Department
800
700
600
500
400
300
200
100
0
Attrition
Yes
No
Sales
Research & Development
Human Resources